

Linear algebra: Exercise MATLAB session

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1. Write a MATLAB code that prompts the user to enter a scalar as the dimension of a vector space and displays the zeros, ones, and elementary vectors in this space.

[Answer.](#)

```
1  clc;
2
3  %Prompt the user to enter a number as the dimension of vector space
4
5  n = input('Enter a number as the dimension of vector space (e.g. 1, 2, ...), n = ');
6
7  %-----
8  %-----Zeros Vector-----
9
10 fprintf('0_%d = \n', n);
11 disp(zeros(n,1));
12
13 %-----
14 %-----Ones Vector-----
15
16 fprintf('1_%d = \n', n);
17 disp(ones(n, 1));
18
19 %-----
20 %-----Elementary Vectors-----
21
22 elementary = eye(n,n);
23
24 for i = 1: n
25     fprintf('e_%d = \n',i);
26     disp(elementary(:,i));
27 end
```

2. Write a MATLAB code that prompts the user to enter a vector and a scalar, then displays the vector, the dimension of vector space containing the vector, the length (norm) of the vector, and the scalar multiplication of the scalar with the vector.

[Answer.](#)

```
1  clc;
2
3  %Prompt the user to enter a vector and a scalar
4
5  user_vector = input('Enter a vector (e.g. [1 2 3]): v = ');
6  user_scalar = input('Enter a Scalar = ');
7
8  %-----
9  %-----Print Vector-----
10
11 fprintf('v = ');
12 disp(user_vector);
13
14 %-----
15 %-----Dimension of Space-----
16
```

```

17 %Notice that in MATLAB length of a vector is defined by the number of its elements, or
    dimension of vector space
18
19 %Initialize the length_vector to zero
20
21 length_vector = 0;
22
23 %Loop through each elements of the vector to count the elements
24
25 for i = user_vector
26     length_vector = length_vector + 1;
27 end
28
29 %%% Another way
30 %%% length_vector = length(user_vector)
31
32 fprintf('The dimension of vector space is: %d\n', length_vector);
33
34 %-----
35 %-----Norm of Vector-----
36
37 %Initialize the norm to zero
38
39 norm_vector = 0;
40
41 %Compute the norm of vector
42
43 for i=1 : length_vector
44     norm_vector = norm_vector + user_vector (i)^2;
45 end
46
47 norm_vector = sqrt(norm_vector);
48
49 %%% Another ways
50 %%% 1) norm_vector = sqrt(sum(user_vector.^2));
51 %%% 2) norm_vector = norm(user_vector);
52
53 fprintf('||v|| = %.2f\n', norm_vector);
54
55 %-----
56 %-----Scalar Multiplication-----
57
58 %Initialize the scalar multiplication vector to zero
59
60 s_multi = zeros(1,length_vector);
61
62 %Compute the norm of vector
63
64 for i=1 : length_vector
65     s_multi(i) = user_scalar * user_vector (i);
66 end
67
68 %%% Another way
69 %%% s_multi = user_scalar * user_vector;
70
71 fprintf('v = ');
72 disp(s_multi);

```

3. Write a MATLAB code that prompts the user to enter two vectors, then displays the addition and the inner product of them, and the angle value between them.

Answer.

```

1 clc;
2
3 % Prompt the user to enter two vectors
4
5 f_vector = input('Enter the first vector a = ');

```

```

6 s_vector = input('Enter the second vector b = ');
7
8 if length(f_vector) == length(s_vector)
9
10     n = length(f_vector);
11
12     %-----
13     %-----ADDITION-----
14
15     %Initialize an element-wise addition vector to zero
16
17     c = zeros(1, n);
18
19     % Loop through each element of the two vectors to compute the addition
20
21     for i = 1:n
22         \hspace*{1cm} c(i) = f_vector(i) + s_vector(i);
23     end
24
25     %%% Another way:
26     %%% c = f_vector + s_vector;
27
28     fprintf('\n')
29     fprintf('a + b = ');
30     disp(c);
31
32     %-----
33     %-----INNER PRODUCT-----
34
35     %Initialize the inner product to zero
36
37     inner_product = 0;
38
39     %Loop through each element of the two vectors to compute the inner product
40
41     for i = 1:n
42         inner_product = inner_product + f_vector(i) * s_vector(i);
43     end
44
45     %%% Another ways:
46     %%% inner_product = sum(f_vector .* s_vector);
47     %%% dot(f_vector, s_vector),
48
49     fprintf('a.b = %.2f\n\n', inner_product);
50
51     %-----
52     %-----Angle Value-----
53
54     %Check if either vector is a zero vector
55
56     if norm(f_vector) == 0 || norm(s_vector) == 0
57         disp('One of the vectors is zero, so it is not possible to find the angle value.
58             ');
59     else
60
61         %Compute the angle value between the two vectors
62
63         angle_value = rad2deg(acos(inner_product / (norm(f_vector) * norm(s_vector))));
64
65         fprintf(' = %.2f\n', angle_value);
66     end
67 else
68     disp('ERROR, these vectors are not in the same vector spaces. ');
69 end

```

4. Write a MATLAB code that prompts the user to enter a vector whose elements are the entries of a diagonal matrix, then displays the matrix and the zeros and identity matrices with the size of diagonal matrix.

Answer.

```
1  clc;
2
3  %-----
4  %-----Diagonal Matrix-----
5
6  %Prompt the user to enter the diagonal entries as a vector
7
8  d = input('Enter the diagonal entries as a vector (e.g [1 2 3]) d = ');
9
10 %Initialize diagonal matrix to zero
11
12 D = zeros(length(d),length(d));
13
14 %Compute diagonal matrix using diagonal vector
15
16 for i = 1 : length(d)
17     D(i,i) = d(i);
18 end
19
20 %%% Anothe way
21 %%% D = diag(d);
22
23 disp('The diagonal matrix is: D = ');
24 disp(D);
25
26 %-----
27 %-----Zeros, Ones and Identity Matrices-----
28
29 fprintf('0_%d = \n',length(d));
30 disp(zeros(size(D)));
31
32 fprintf('1_%d = \n',length(d));
33 disp(ones(size(D)));
34
35 fprintf('I_%d = \n',length(d));
36 disp(eye(size(D)));
```

5. Write a MATLAB code that prompts the user to enter a matrix, then displays the matrix and the size, transpose, and trace of matrix.

Answer.

```
1  clc;
2
3  %Prompt the user to enter a matrix
4
5  user_matrix = input('Enter a matrix (e.g [1 2 3; 4 5 6]): A = ');
6
7  %-----
8  %-----Print Matrix-----
9
10 disp('A = ');
11 disp(user_matrix);
12
13 %-----
14 %-----Size of Matrix-----
15
16 size_matrix = size(user_matrix);
17 fprintf('The size of matrix is: %d * %d\n', size_matrix(1), size_matrix(2));
18
19 %-----
20 %-----Transpose of Matrix-----
21
22 %Compute the size of matrix transpose
23
```

```

24 size_tmatrix = zeros(1,2);
25 size_tmatrix(1) = size_matrix(2);
26 size_tmatrix(2) = size_matrix(1);
27
28 %Initialize matrix transpose to zero
29
30 t_matrix = zeros(size_tmatrix);
31
32 %Compute matrix transpose
33
34 for i = 1 : size_tmatrix(1)
35     for j = 1 : size_tmatrix(2)
36         t_matrix(i,j) = user_matrix(j,i);
37     end
38 end
39
40 %%% Another way
41 %%% t_matrix = transpose(user_matrix);
42
43 disp('A^T = ');
44 disp(t_matrix);
45
46 %-----
47 %-----Trace of Matrix-----
48
49 %Initialize the tace to zero
50
51 trace_matrix = 0;
52
53 %Compute the trace of matrix
54
55 if size_matrix(1) == size_matrix(2)
56
57     for i = 1 : size_matrix(1)
58         trace_matrix = trace_matrix + user_matrix(i,i);
59     end
60
61     %%% Another ways
62     %%% 1) sum(diag(A))
63     %%% 2) t_matrix = trace(user_matrix);
64
65     fprintf('tr(A) = %d\n', trace_matrix);
66
67 else
68     disp('It is not possible to compute trace of matrix.');
```

6. Write a MATLAB code that prompts the user to enter a matrix, a vector, and a scalar, then displays the matrix, the vector, and the scalar-matrix and vector-matrix multiplications.

Answer.

```

1  clc;
2
3  %Prompt the user to enter a matrix, a vector, and a scalar
4
5  user_matrix = input('Enter a vector (e.g [1 2 3; 4 5 6]): A = ');
6  user_vector = input('Enter a vector (e.g [1; 2; 3]): v = ');
7  user_scalar = input('Enter a Scalar = ');
8
9  %-----
10 %-----Print Matrix and Vector-----
11
12 disp('A =');
13 disp(user_matrix);
14
15 disp('v =');
16 disp(user_vector);
```

```

17
18 %Compute the size of matrix
19
20 size_matrix = size(user_matrix);
21
22 %-----
23 %-----Scalar Multiplication-----
24
25 %Initialize the scalar multiplication to zero
26
27 s_multi = zeros(size_matrix);
28
29 %Compute the the scalar multiplication
30
31 for i=1 : size_matrix(1)
32
33     for j=1 : size_matrix(2)
34         s_multi(i,j) = user_scalar * user_matrix (i,j);
35     end
36
37 end
38
39 %%% Another ways
40 %%% 1) for i=1 : size_matrix(1)
41 %%%     s_multi(i,:) = user_scalar * user_matrix (i,:);
42 %%%     end
43 %%% 2) s_multi = a * user_matrix;
44
45 disp('A = ');
46 disp(s_multi);
47
48 %-----
49 %-----Matrix-Vector Multiplication-----
50
51 %Initialize the matrix-vector multiplication to zero
52
53 v_multi = zeros(size_matrix(1),1);
54
55 if length(user_vector) == size_matrix(2)
56
57     %Compute the the vector multiplication Av
58
59     for i=1 : size_matrix(1)
60         v_multi(i) = dot(user_matrix(i,:),user_vector);
61     end
62
63     %%% Another way
64     %%% v_multi = user_matrix * user_vector;
65
66     disp('Av = ');
67     disp(v_multi);
68 else
69     disp('ERROR, it is not possible to compute Av.');
```

```

70 end

```

7. Write a MATLAB code that prompts the user to enter two matrices, then displays addition of matrices and the matrix-matrix multiplication of them.

Answer.

```

1 clc;
2
3 % Prompt the user to enter two matrices
4
5 f_matrix = input('Enter the first matrix A = ');
6 s_matrix = input('Enter the second matrix B = ');
7
8 size_fmatrix = size(f_matrix);

```

```

9 size_smatrix = size(s_matrix);
10
11 %-----
12 %-----ADDITION-----
13
14 if size_fmatrix == size_smatrix
15
16     % Initialize an element-wise addition matrix to zero
17
18     C = zeros(size_fmatrix);
19
20     % Loop through each element of the two matrices to compute
21
22     for i = 1:size_fmatrix(1)
23         for j = 1:size_fmatrix(2)
24             C(i,j) = f_matrix(i,j) + s_matrix(i,j);
25         end
26     end
27
28     %%% Another way:
29     %%% add = f_matrix + s_matrix;
30
31     fprintf('\n')
32     disp('A + B = ');
33     disp(C);
34
35 else
36     disp(['These matrices are not in the same vector spaces,' ...
37         ' so it is not possible to compute A + B.']);
38 end
39
40 %-----
41 %-----Matrix-Matrix Multiplication-----
42
43 if size_fmatrix(2) == size_smatrix(1)
44
45     % Initialize the matrix-matrix multiplication to zero
46     D = zeros(size_fmatrix(1),size_smatrix(2));
47
48     % Loop through each element of the two vectors to compute the inner product
49     for i = 1 : size_fmatrix(1)
50         for j = 1 : size_smatrix(2)
51             D(i,j) = dot(f_matrix(i,:),s_matrix(:,j));
52         end
53     end
54
55     %%% Another way:
56     %%% D = f_matrix * s_matrix;
57
58     disp('AB = ');
59     disp(D);
60
61 else
62     disp(['The number of columns of A is not equal to the number of rows of B,' ...
63         ' so it is not possible to compute AB.']);
64 end

```